EUROPEAN COOPERATION IN SCIENCE & TECHNOLOGY (COST)

Report on the outcomes of a presentation and participation

Young Researcher and Innovator Conference Grant (YRICG)

Action Number:	**CA24153**	**Grant Reference:**	E-COST-GRANT-CA24153-b2bf4e39
Grantee Name:	**Mr MAAMAR ARBOUZ**	**Awarded Amount:**	**EUR 1,400.00** (Approved at 100%)
Home Institution:	University of Khemis Miliana (Algeria)	**Attendance Dates:**	12/10/2026 – 15/10/2026
Attendance Type:	Face to face (In-person)	**Host Country:**	Italy (IT)

1. Conference Details

Conference Title:	**2nd European Workshop on Artificial Nighttime Light Remote Sensing**
Conference Web-page:	https://nikal.eventsair.com/optical-nighttime-lights-workshop
Conference Venue:	European Space Agency (ESA-ESRIN), Largo Galileo Galilei 1, 00044 Frascati (Rome), Italy

2. Accepted Contribution Details

Title of Presentation:	**Spatio-Temporal Graph Neural Networks for Multi-Source Disaggregation and Environmental Impact Assessment of Artificial Nighttime Light Remote Sensing**
Type of Presentation:	Oral / Poster Presentation
Co-authors:	Maamar Arbouz (Lead Author & Investigator)
Session / Track:	Advanced Remote Sensing Methodologies, Multi-Sensor Radiometric Fusion, and Environmental & Ecological Impacts of Artificial Light at Night (ALAN)

3. Outcome of the Conference Participation (Scientific Output, Visibility & Future Collaborations — 348 words)

The participation in the 2nd European Workshop on Artificial Nighttime Light Remote Sensing at ESA-ESRIN provided an invaluable international platform to present my research on Spatio-Temporal Graph Neural Networks (ST-GNN) for nocturnal radiance disaggregation. My presentation focused on resolving the fundamental trade-off in spaceborne nighttime light (NTL) observation between temporal revisit and spatial resolution, specifically downscaling 500-meter VIIRS-DNB nocturnal radiance to a 10-meter grid by integrating Sentinel-2 spectral indices (NDBI, NDVI) and structural road geometries with guaranteed radiometric energy conservation.

The presentation generated high engagement and constructive technical dialogue with leading Earth Observation scientists, mission specialists, and optical instrumentation engineers from ESA, CNR, and partner European academic institutions. Key technical discussions centered on the mathematical formulation of spatial graph attention mechanisms, cross-sensor radiometric normalization across heterogeneous atmospheric profiles, and the operational application of super-resolved nocturnal radiance for delineating skyglow intrusion into fragile Mediterranean ecosystems and monitoring critical infrastructure resilience during localized grid disruptions.

In terms of academic visibility, attending this flagship event as an early-career researcher from the University of Khemis Miliana (Algeria) significantly enhanced my international profile within the European remote sensing and spatial data science communities. It highlighted the contribution of North African researchers to cutting-edge geometric deep learning applied to satellite Earth observation.

Crucially, the conference enabled the establishment of promising bilateral and multilateral contacts for future scientific collaboration. Constructive exchange took place with research groups specializing in nocturnal environmental monitoring, atmospheric radiative transfer modeling, and urban dynamics from institutions across Italy, Poland, and the United Kingdom. Discussions were initiated regarding joint validation campaigns using co-registered airborne and ground photometer datasets, potential co-authorship on peer-reviewed journal papers extending the ST-GNN framework to multi-temporal planetary datasets, and prospective proposals for Short-Term Scientific Missions (STSM) and collaborative consortia under European Horizon initiatives. Furthermore, direct interaction with ESA-ESRIN personnel provided strategic insights into upcoming mission requirements. I express my sincere gratitude to the COST Association

and Action CA24153 for awarding this conference grant.

4. Acknowledgement of Inclusion of Necessary Supporting Documents

I confirm that the following required documents have been uploaded on e-COST as an integral part of this report:

- **The certificate of conference attendance / acceptance** issued by the conference organiser (ESA Conference Bureau, ESA-ESRIN).
- **The programme of the conference or book of abstracts / proceedings** indicating the presentation of the grantee.

Date: 19 October 2026	**Grantee Signature:**	**MAAMAR ARBOUZ (Electronically Signed)**